UAPOML

Q1.

(a) $R_1 = \dfrac{108-100}{100} = \dfrac{8}{100} = 0.08$

$R_3 = \dfrac{115-103}{103} = \dfrac{12}{103} \approx 0.1165$

$R_2 = \dfrac{103-108}{108} = \dfrac{-5}{108} \approx -0.0463$

$R_4 = \dfrac{110-115}{115} = \dfrac{-5}{115} \approx -0.0435$

(b) Log Returns

$r_1 = \ln\left(\dfrac{108}{100}\right) \approx 0.0770$

$r_3 = \ln\left(\dfrac{115}{103}\right) \approx 0.1102$

$r_2 = \ln\left(\dfrac{103}{108}\right) \approx -0.0474$

$r_4 = \ln\left(\dfrac{110}{115}\right) \approx -0.0445$

(c) for small returns,

$$\ln(1+x) \approx x.$$

Comparision is least accurate in month 4.

Q2.

(a) $E[ax+by] = a\,E[x] + b\,E[y]$

$$E[R_p] = w_1 E[R_1] + w_2 E[R_2]$$
$$= (0.4)(0.15) + (0.6)(0.07)$$
$$= 0.102$$

$$\boxed{E[R_p] = 10.2\%}$$

(b) No, the Expected return does not change if R_1 & R_2 are correlated.

→ Expectation depends only on the weighted avg. of individual expected returns.

→ Correlation affects the portfolio variance / risk, not the expected return.

(Q3) $Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)$

$X = w_1 R_1 \qquad Y = w_2 R_2$

$Var(R_p) = Var(w_1 R_1 + w_2 R_2)$

$= Var(w_1 R_1) + Var(w_2 R_2) + 2\,Cov(w_1 R_1, w_2 R_2)$

$= w_1^2 Var(R_1) + w_2^2 Var(R_2) + 2 w_1 w_2\, Cov(R_1, R_2)$

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2\, cov(R_1, R_2)$

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

(Q4)

$\sigma_p^2 = (0.5)^2 (0.25)^2 + (0.5)^2 (0.12)^2 + 2(0.5)(0.5)\rho(0.25)(0.12)$

$= 0.25(0.0625) + 0.25(0.0144) + 0.5\rho(0.03)$

$= 0.015625 + 0.0036 + 0.015\rho$

$\sigma_p^2 = 0.019225 + 0.015\rho$

(a)

$\rho = +1$	$\rho = 0$	$\rho = -1$
$\sigma_p^2 = 0.019225 + 0.015$	$\sigma_p^2 = 0.019225$	$\sigma_p^2 = 0.004225$
$= 0.034225$	$\boxed{\sigma_p = 13.87\%}$	$\boxed{\sigma_p = 6.5\%}$
$\sigma_p = \sqrt{0.034225} = 0.185$		
$\boxed{\sigma_p = 18.5\%}$		

(b) $w_A\sigma_A + w_B\sigma_B = (0.5)(0.25) + (0.5)(0.12)$

$= 0.125 + 0.06$

$= 0.185 = \boxed{18.5\%}$

when ($\rho = +1$)

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$$

if $\rho = 1$

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B$$

$$\boxed{\sigma_p = w_A\sigma_A + w_B\sigma_B}$$

(c) $\rho = -1$

$$\sigma_p = |w_A\sigma_A - w_B\sigma_B|$$

To make risk zero,

$$w_A\sigma_A = w_B\sigma_B$$

since, $w_B = 1 - w_A$

we get,

$$w_A\sigma_A = (1-w_A)\sigma_B$$

$$\boxed{w_A^* = \frac{0.12}{0.37} \approx 32.43\%} \qquad \boxed{w_B^* = 1 - w_A^* = 67.57\%}$$

Q5. $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$

If $\rho = 1$, the assets move perfectly together, so there is no diversification benefit. In that case,

$\sigma_p = w_1\sigma_1 + w_2\sigma_2$, which is just the weighted avg. of risks.

However, when $\rho_{12} < 1$, the assets do not move exactly together. Sometimes one asset rises while the other falls or rises by less. This ↓ the cov term in the var formula, making total portfolio variance smaller.

Q6. $\rho_{12} = 0.4$, $\rho_{13} = 0.1$, $\rho_{23} = 0.2$ & $\rho_{ii} = 1$

$$\Sigma_{ij} = \rho_{ij}\, \sigma_i \sigma_j$$

(a)

$\Sigma_{11} = \sigma_1^2 = 0.0625$

$\Sigma_{22} = \sigma_2^2 = 0.0144$

$\Sigma_{33} = \sigma_3^2 = 0.0016$

$\Sigma_{12} = \rho_{12}\sigma_1\sigma_2 = 0.012$

$\Sigma_{13} = \rho_{13}\sigma_1\sigma_3 = 0.001$

$\Sigma_{23} = \rho_{23}\sigma_2\sigma_3 = 0.00096$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix}$$

(b) $\Sigma_{12} = \Sigma_{21}$; $\Sigma_{23} = \Sigma_{32}$; $\Sigma_{13} = \Sigma_{31}$

Hence $\boxed{\Sigma = \Sigma^T}$ symmetric matrix.

Q7.

(a)

X

$S_X = \dfrac{0.14 - 0.04}{0.22} \approx 0.455$

Y

$S_Y = \dfrac{0.09 - 0.04}{0.10} \approx 0.50$

Z

$S_Z = \dfrac{0.20 - 0.04}{0.35} \approx 0.457$

(b) Y > Z > X

(c) Although portfolio Z has the highest expected return (20%), it also has a much higher risk (35%).

Y provides:

→ lower volatility
→ better risk-adjusted performance
→ and a higher Sharpe-ratio

Y > Z

Q6.

(a) It is the portfolio on the feasible set that has the smallest possible variance (or standard deviation) among all risky portfolios.

It is called the leftmost point on the min variance frontier because:

- The horizontal axis measures portfolio risk (σ_p).
- moving left means lower risk,
- and no portfolio can exist with risk smaller than the GMV portfolio.

(b) No, the lower half is inefficient b/c it gives lower return for the same level of risk. Rational investors always choose portfolios on the upper efficient frontier.

(c) CML is drawn from the risk-free rate tangent to the efficient frontier at the market portfolio.

Its slope equals: $\frac{E[R_m] - R_f}{\sigma_m}$ which represents the extra return earned per unit of risk.

Q9.

(a) $SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{T}} = \frac{0.30}{\sqrt{36}} = 0.05$

$SE = 5\%$

(b) MVO has an error maximisation problem because small estimation errors in expected returns can lead to very extreme portfolio weights.

The optimiser tends to over-allocate to assets with slightly overestimated returns.

(c) Unique parameters = N variances + $\frac{N(N-1)}{2}$ unique cov

$= 50 + \frac{50 \times 49}{2} = 1275$ unique parameter.

X X X